Lab -8: Experiment No: 8

AIM

Configuring a Cryptocurrency Wallet using Metamask and integrating it with a DApp

THEORY AND BACKGROUND

1.Cryptocurrency Wallet

A cryptocurrency wallet is a software application or digital tool used to manage blockchain accounts and interact with blockchain networks.

A wallet does not physically store cryptocurrency. Instead, it manages the **private keys** that allow a user to access and authorize transactions associated with their blockchain address.

A typical Ethereum wallet provides:

- Ethereum wallet address
- Private key management
- Account balance
- Transaction signing
- Connection with decentralized applications
- Interaction with smart contracts

2.MetaMask

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with Ethereum-compatible blockchain networks and decentralized applications.

It acts as a bridge between:

User → MetaMask → Blockchain / Smart Contract → DApp

MetaMask provides the following functionality:

1. Creates and manages Ethereum accounts.
2. Stores and manages cryptographic keys.
3. Displays wallet balances.
4. Allows users to approve blockchain transactions.
5. Allows DApps to request access to the user's wallet.
6. Signs transactions before they are submitted to the blockchain.

Supports Ethereum test networks such as **Sepolia**.

3. **Ethereum Network**

Ethereum is a blockchain platform that supports programmable transactions through **smart contracts**.

For development and testing, a test network such as **Sepolia** can be used instead of the Ethereum mainnet.

The advantage of using a test network is that developers can test:

- Wallet connections
- Smart contracts
- Transactions
- DApp functionality without using real cryptocurrency.

4.Ethereum Wallet Address

Every Ethereum account has a unique wallet address used to identify the account on the blockchain.

The address can be used to:

- Receive ETH
- Send ETH
- Interact with smart contracts
- Identify the account performing a blockchain transaction

When a DApp connects to MetaMask, it can obtain the user's wallet address after the user grants permission.

5.DApp

A **Decentralized Application (DApp)** is an application that interacts with blockchain-based infrastructure rather than relying completely on a centralized backend.

A typical DApp consists of:

Frontend → Web3 provider / Wallet → Smart Contract → Blockchain

In the Ethlance implementation:

The frontend provides the user interface.

MetaMask provides wallet connectivity.

Ethereum Sepolia provides the blockchain network.

Smart contracts provide blockchain-based functionality.

Transactions are authorized using the user's wallet.

6.Wallet–DApp Integration

The basic process of integrating MetaMask with a DApp is:

1. Open the DApp in a browser.
2. Install/open MetaMask.
3. Select the required Ethereum network.
4. Select the required wallet account.
5. Click **Connect Wallet** on the DApp.
6. MetaMask displays a connection request.
7. User approves the request.
8. The DApp obtains access to the wallet address.
9. When a blockchain operation is performed, the DApp requests a transaction through MetaMask.
10. The user reviews and confirms the transaction.
11. MetaMask signs and submits the transaction to the blockchain.
12. The transaction is recorded on the blockchain.

7.Gas Fees

Blockchain transactions require computational resources from the Ethereum network. The associated transaction cost is called **gas**.

Gas is required for operations such as:

- Deploying smart contracts
- Calling functions that modify blockchain state
- Transferring ETH
- Performing other blockchain transactions

In a test network such as Sepolia, test ETH can be obtained from a faucet and used to perform these operations without spending real ETH.

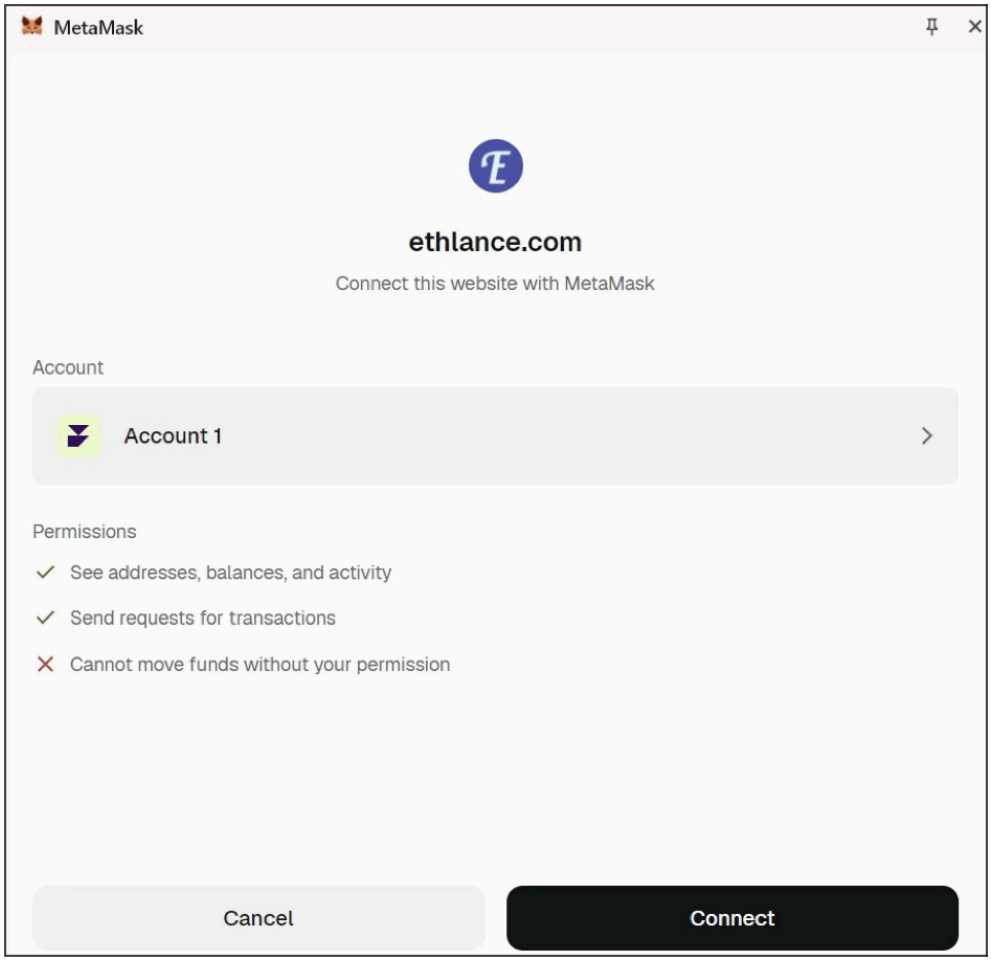
8.Smart Contracts

A smart contract is a program deployed on a blockchain that automatically executes predefined logic when its conditions are satisfied.

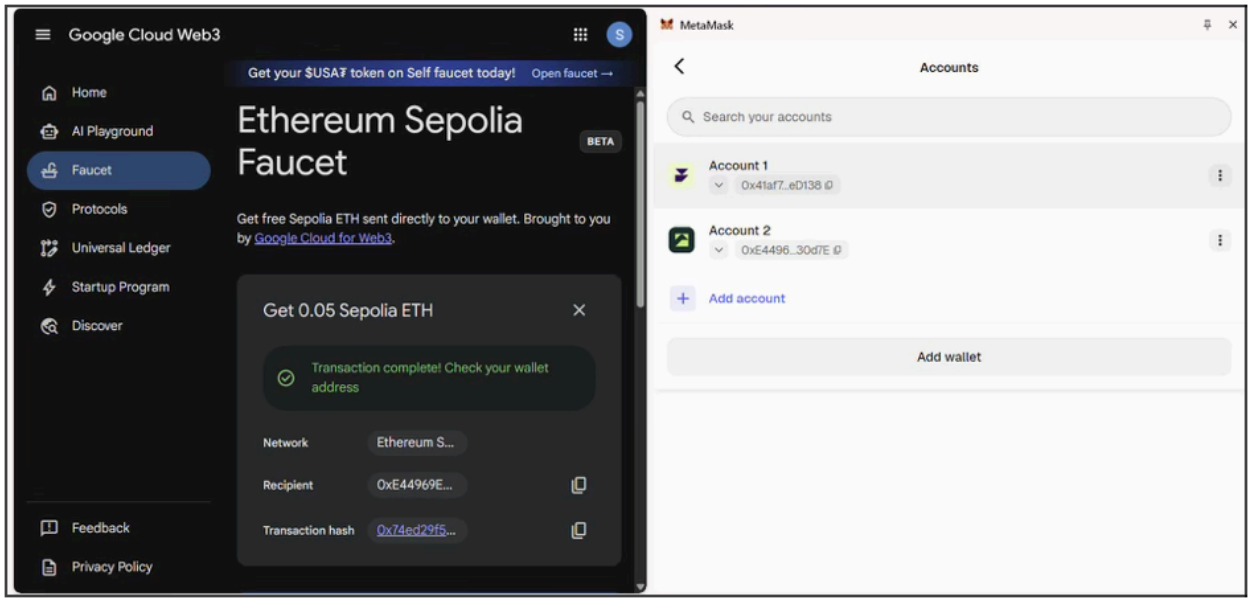
In the Ethlance DApp, smart contracts are used to implement functionality related to the decentralized freelance marketplace.

Procedure:

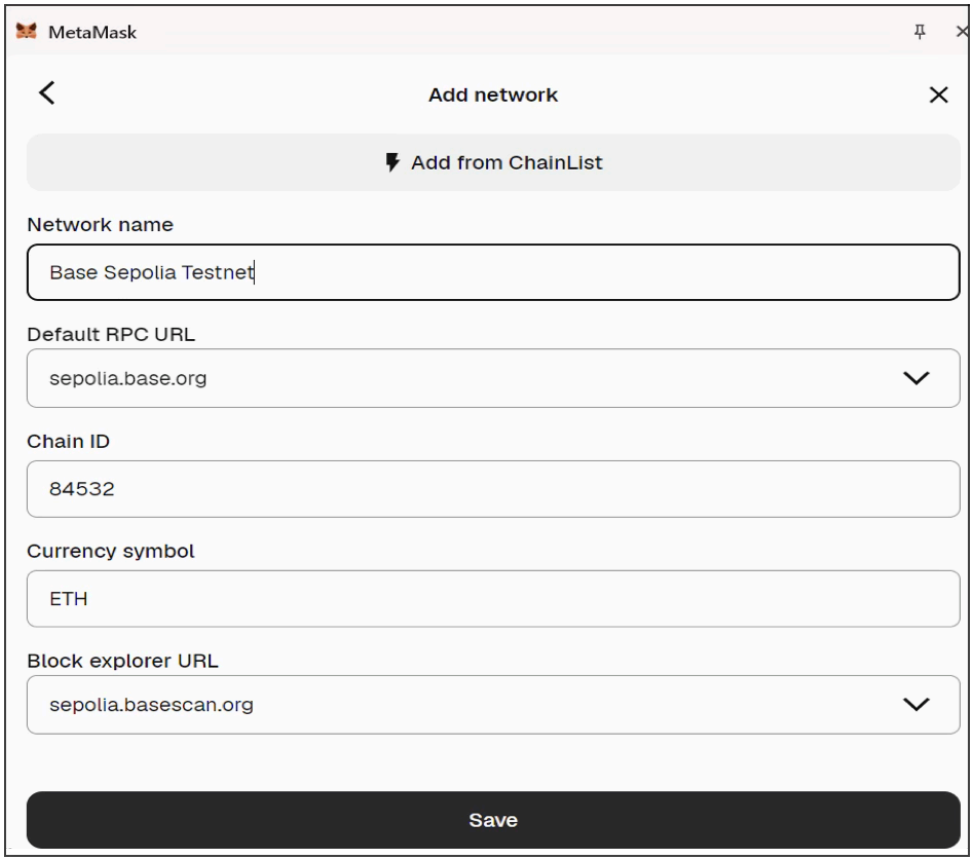
1. Opened/installed the **MetaMask** browser extension.



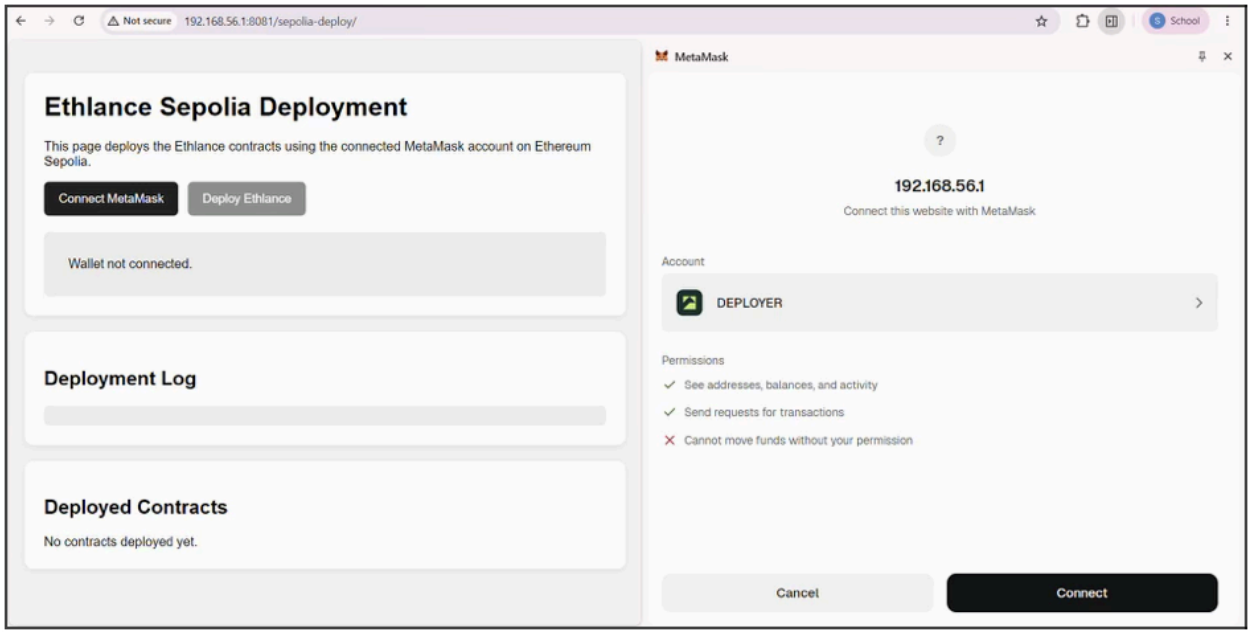
2. Configured an Ethereum wallet account in MetaMask.



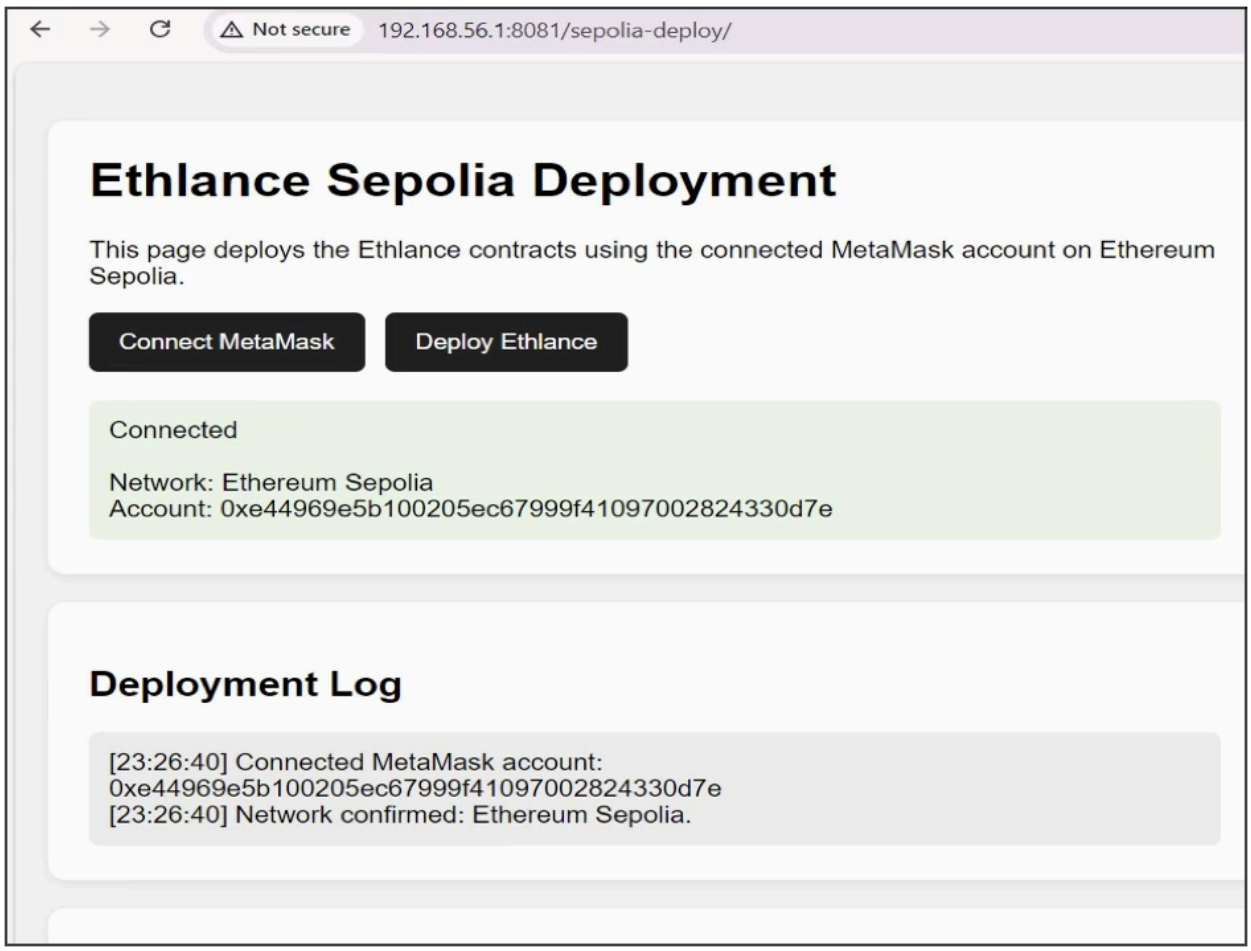
3. Added/configured the **Ethereum Sepolia Test Network**.



4. Obtained test ETH from the Sepolia faucet.

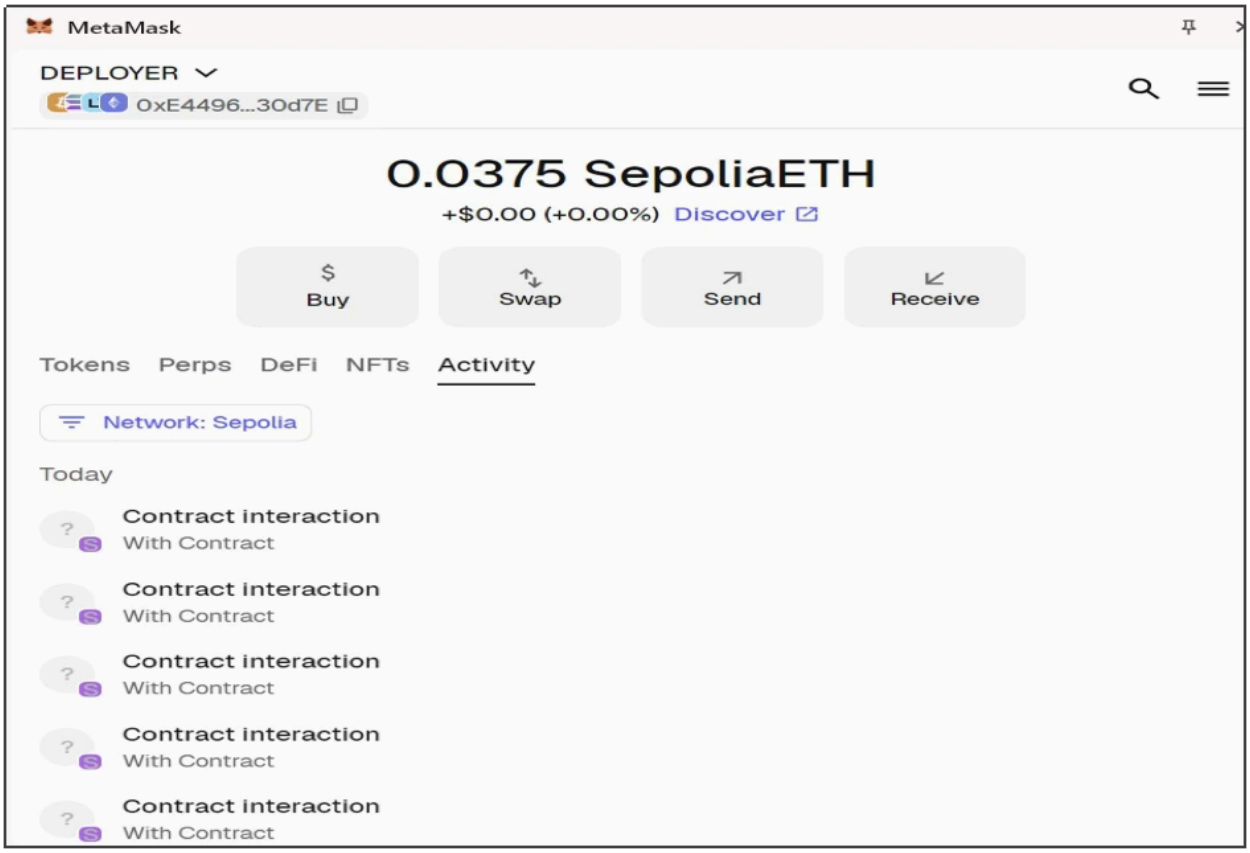


5. Verified the test ETH balance in MetaMask.

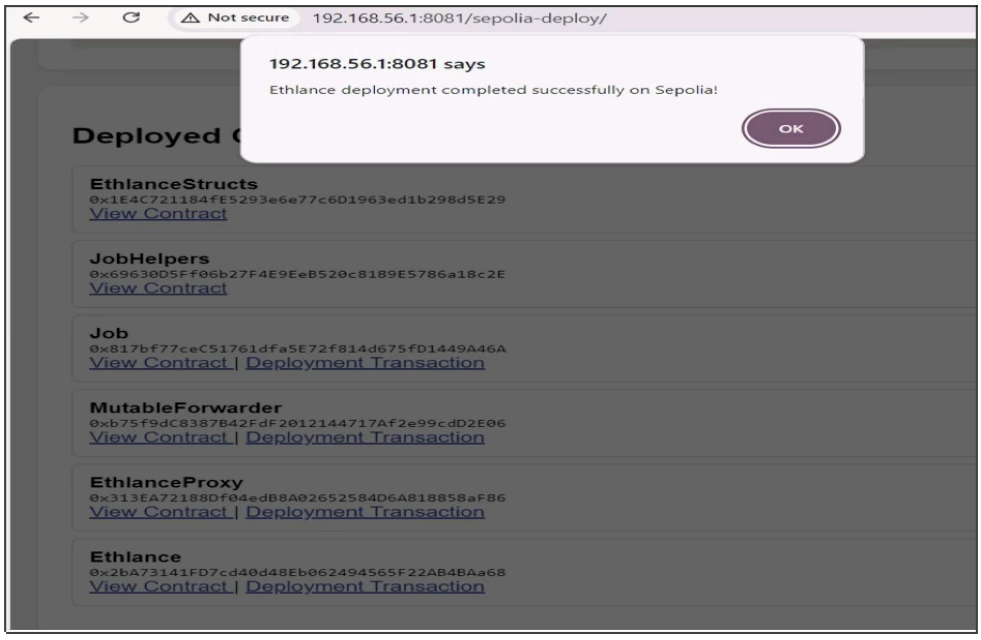


6. Opened the Ethlance DApp.

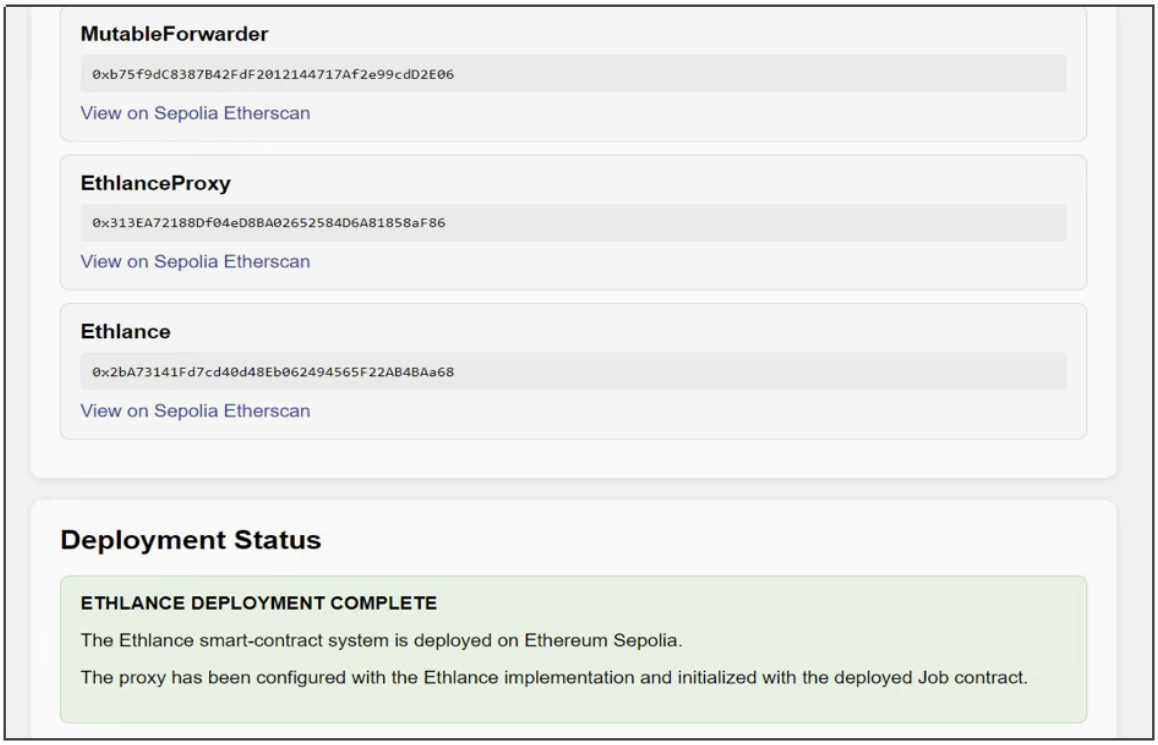
7. Selected the option to connect the cryptocurrency wallet.
8. MetaMask displayed a permission request to connect the account.



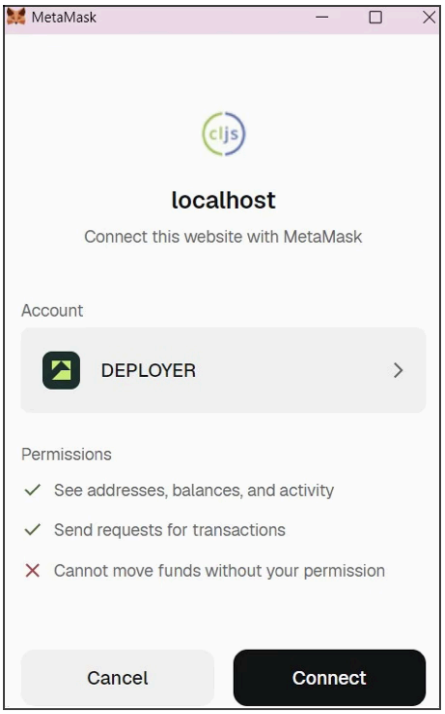
9. Approved the connection request.
10. The Ethlance DApp was connected to the MetaMask wallet.
11. The Ethlance implementation was deployed/interacted with on the Sepolia network.

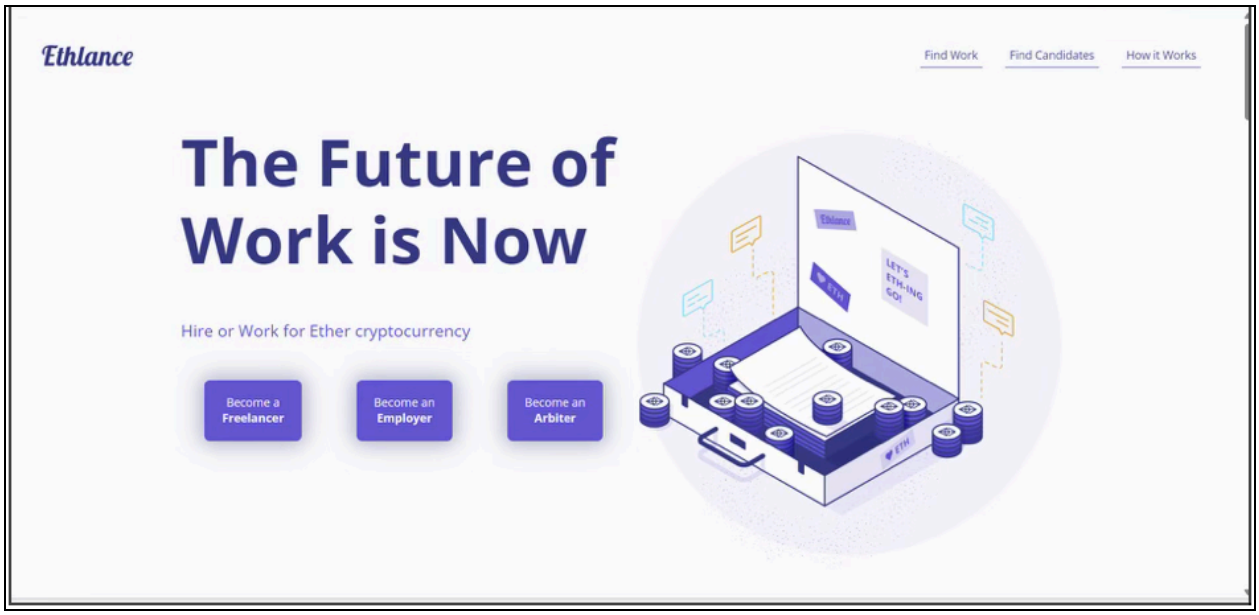


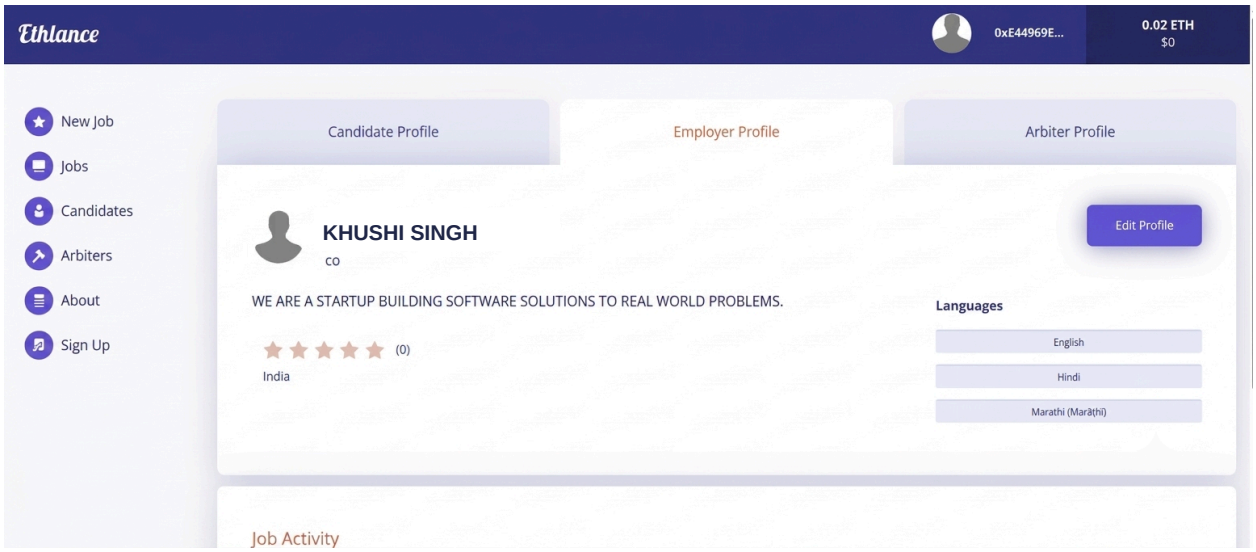
12. MetaMask was used to approve blockchain transactions.
13. The deployed DApp was accessed locally and the connected wallet was used to perform DApp operations.



14. Verified that the DApp could interact with the blockchain through the configured MetaMask account.







CONCLUSION:

Thus, the MetaMask wallet was configured for use with the Ethereum Sepolia test network. Test ETH was obtained for practice, and the Ethlance DApp was connected to the wallet. This activity demonstrated how a wallet can provide account access and authorize blockchain interactions through a decentralized application.